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An introduction to Bayesian credible intervals for sampling error in DNA profiles

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Experts presenting statistical DNA evidence are often challenged about their ability to use databases of only a few hundred people to claim probabilities in the billions. This challenge is countered with the inclusion of a sampling error calculation in the case report. Sampling error is present in every DNA calculation because the statistics rely on a sample (database) of individuals to estimate population frequencies of alleles and this error must be quantified. This paper explains in simple terms how Bayesian estimation theory is used to produce Bayesian credible intervals for sampling error calculations in DNA profiles.

Keywords: sampling error; DNA evidence; Bayesian credible intervals.

1. Introduction

One of the most common questions asked of an expert presenting statistical evidence in a case involving DNA is, "your database only has 200 people in it—how can you present statistics with probabilities of 1 in 1 billion?" That is simply an example using hypothetical numbers, but a database of 200 is consistent with the sizes of the databases that the Federal Bureau of Investigation (FBI) provides (Budowle & Moretti, 1999) for statistical calculation in the United States. A well-prepared analyst can answer this question by providing an assessment of sampling error in the calculations.

When DNA evidence is presented to the court, it is typically characterized by some numerical measure of strength. This may be in the form of a match probability, a likelihood ratio (LR) or perhaps a paternity index. All of these measures have an associated amount of sampling error. Sampling error in this context is statistical sampling error, not errors in the method of DNA sampling. Sampling error is utterly unavoidable as it occurs through the simple process of taking a sample. The reader may wonder at this point where or what is the sample in the DNA context? All statistical measures of the weight of the DNA evidence rely on one or more sets of allele frequencies. Ideally, these allele frequencies would be allele probabilities, known exactly. Of course this is unlikely to ever happen. In practice, the allele frequencies come from a database collected by the forensic agency associated with the case or from a set of published frequencies, such as those provided by the FBI. The frequencies from the database are estimates of the true allele probabilities. Therefore, the databases are samples and the allele frequencies are estimates which inherently have sampling error. The practical implication here is that if a different database were to be collected, it is more than likely that it would

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give different allele frequencies and hence have a different match probability or LR. The pertinent question is, "how different could the frequencies be and what effect will this have on the LR?" In order to answer this question it is necessary to try and quantify the uncertainty about the allele frequencies. This uncertainty is often expressed in statistical literature as a confidence interval—one part of which is expressed as the margin of error in political polls, for example. Simply put, without the usual philosophical statements, a confidence interval is a frequentist measurement—meaning that its interpretation involves long-term averages. The Bayesian counterparts are the credible interval or sometimes the region of highest posterior density (HPD). For the applications described here, they can be regarded as being equivalent.

Unfortunately, measures of sampling error are not commonly presented in court. This omission probably has more to do with a lack of understanding than any willful desire to misguide the court. In cases where sampling error is presented, the measures used are either ad hoc (such as the factor of 10 rule) or based on results which have been shown to be wrong (such as the size bias correction). It may seem unimportant to present sampling error calculations when the LRs (for modern single contributor DNA cases) are routinely in the order of 10^{14}. In this example, the resulting lower bound may "only" be in the order 10^{12}. While this lower bound is unlikely to have any effect on the decision processes of the jury, it is the belief of this author that presenting the calculation with error is more "scientifically honest". Furthermore, demonstrating that these errors have been calculated is the only correct and convincing response to the almost predictable questions about the size of the database and the validity of the results. The most appropriate comment on the subject perhaps comes from Buckleton (2004) "An analyst who is prepared for such a cross examination will definitely present better evidence to the court than one who chooses to answer, 'would it make any difference?'"

Several authors have addressed the problem of sampling error. There have been several methods suggested to characterize the sampling error of an estimated match probability, generally through the use of confidence intervals. The first National Research Council (NRC) report on forensic DNA evidence (NRC, 1992) suggested that the frequency of each constituent allele be replaced by a binomial upper confidence limit and these limits then be multiplied across alleles. Another simplistic approach, with some empirical support, was to multiply and divide the estimated match probability by a factor of 10, suggested by the second NRC report on forensic DNA evidence (NRC, 1996). Weir (1996) suggested a bootstrapping approach by resampling individuals in some population sample. Chakraborty *et al.* (1993) appealed to the asymptotic normality of the logarithm of the product of allele frequencies to provide an analytical expression for the confidence interval; this was extended by the NRC (1996) to allow for population structure. Balding (1995) gave both a likelihood support interval and a 'conservative Bayesian estimate'. Curran *et al.* (2002) gave a method based on a Bayesian approach. Buckleton and Curran (2004) showed how this method is highly flexible and perhaps more theoretically sound than other approaches. It is the author's belief that this method has met with some resistance because of a misperception that it is unduly complex. The hope is that this paper will dispel some of these notions.

The topic of Bayesian estimation as opposed to Bayesian reasoning is explained in Section 2. Section 3 deals with the choice of prior distributions and attempts to dispense with some common criticisms. Section 4 explains the concept of Bayesian credible intervals and compares them to the more familiar confidence intervals. Section 5 returns to the problem of assessing sampling error in an LR for a case involving DNA evidence. Section 6 explains briefly how to extend the results to multiple alleles and loci and Section 7 attempts to draw some conclusions.

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2. Bayes Theorem and Bayesian estimation

Individuals familiar with the statistical assessment of forensic evidence in a Bayesian framework are accustomed to the odds form of Bayes Theorem to show how the LR can be combined with the prior odds (on guilt[1]) to give the posterior odds (on guilt). However, this is not how most statisticians use Bayes Theorem although the thought process is similar. In Bayesian estimation, the focus is on the value of a parameter such as a mean or a proportion. In any framework, Bayesian or other, the parameter may be estimated from the data. However, if the goal is to make direct inference about the possible values that the parameter may take with respect to the collected data, then some Bayesian methodology is necessary. More specifically, the goal is to derive the posterior distribution of the parameter given the data. A probability distribution (or density) tells a statistician such things as where the most probable values of the parameter lie and the likely range of values that a parameter may take on. Once this distribution has been obtained, standard quantities such as the mean and variance can be calculated. The posterior distribution is constructed by combining a prior distribution (belief) about the value of the parameter with the likelihood of the parameter given the data that has been observed.

The way in which the prior and likelihood combine to give the posterior is given by Bayes Theorem

$$\Pr(\theta|\text{data}) \propto \Pr(\text{data}|\theta)\Pr(\theta). \tag{1}$$

In words, (1) gives "the probability of the parameter given the data is proportional to the probability of the data given the parameter times the probability of the parameter". The reader should not be confused by θ. It simply is a placeholder for the parameters of interest. For example, θ could be a population mean μ, or a population proportion π. The 'scaling' factor that must be calculated in order to calculate the posterior probability exactly is given in Appendix 1. This formulation is not too different from the traditional LR situation. The scientist/caseworker has some belief about a parameter (maybe no idea at all), an experiment is performed and some data are collected to gain some information about the parameter, and then this information is combined with the prior belief to get an updated belief about the parameter. So the equation can be broken down into the posterior, $\Pr(\theta|\text{data})$, the likelihood, $\Pr(\text{data}|\theta)$ and the prior $\Pr(\theta)$, i.e. (1) may be expressed as

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}. \tag{2}$$

Some readers may have noticed there has been a switch from talking about distributions (or probability density functions (pdfs)) to talking about probabilities (actually cumulative distribution functions (cdfs)). Bayes Theorem may be applied both to probabilities and pdfs.

It should also be noted at this point that there is absolutely no controversy about the mathematical definition of Bayes Theorem. There is a misconception that Bayes Theorem is controversial. However, it is only the application of Bayes Theorem that is debated, not the theorem itself, which is a simple statement of fact easily verified by mathematics.

Several terms were used in the definition of expression (2). The likelihood function is usually dictated by the model used for the problem. The prior distributions placed on model parameters can be a different matter.

[1] We use the simplistic concept of guilt here to represent a variety of hypotheses such as the suspect is a contributor to the crime scene stain or the suspect was in contact with the crime scene.

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3. Prior probabilities

The easiest way to explain the concept of prior distributions/probabilities is in the context of a simple example. Assume that a forensic scientist wishes to assess the frequency of various alleles at a locus. Assume further that the locus of interest has only two alleles, labelled A and B. An example of a locus with such properties is the low density lipoprotein receptor (LDLR) locus available in the Perkin Elmer AmpliType® PM (PolyMarker) kit. Given that individuals can have only A or B alleles (null alleles and other complications are being ignored here) then $\Pr(\mathrm{A}) + \Pr(\mathrm{B}) = 1$, or $\Pr(\mathrm{B}) = 1 - \Pr(\mathrm{A})$. Therefore, in this case, it suffices to estimate the probability of allele A. Let this value be denoted by π_{A}.

The question then becomes one of estimating the probability of allele A in the population. Traditionally, a sample of people is taken (a database), who are then typed and the number of A alleles is counted. That is, the estimate of the probability is the sample frequency

$$f_{\mathrm{A}} = \hat{\pi}_{\mathrm{A}} = \frac{\text{No. of A's}}{2N}, \tag{3}$$

where N is the number of individuals in the database. For example, say a sample of 10 people was selected and in that sample seven alleles of type A were observed. The estimate of the probability of allele A in the population is $7/20 = 0.35$. At this point the reader should think about the following question: "what was known about the probability of allele A, π_{A}, before the sample was taken?" Any answer to this question can be regarded as an assessment of the prior probability or belief about π_{A}. Furthermore, it should be clear that a formal method of expressing this 'prior knowledge' is necessary and that method should be flexible enough to represent any set of beliefs about π_{A}. Traditionally, this is done by making probability statements about the values that π_{A} may assume. For example, one might say that there is a 10% chance that π_{A} is less than 0.1, a 10% chance that π_{A} is greater than 0.9 and an 80% chance that π_{A} is between 0.1 and 0.9. Together, these probabilities add up to one and what has been described is called a cdf. The cdfs describe the area under a curve of the pdf. The key fact is that the area (not the height) of the curve measures probability. This is best shown by the graphs in Fig. 1.

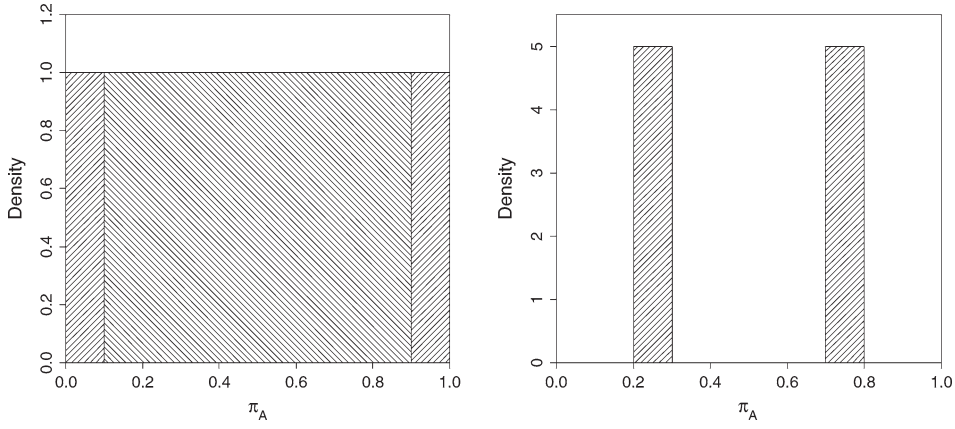


FIG. 1. Possible prior distribution functions for π_A.

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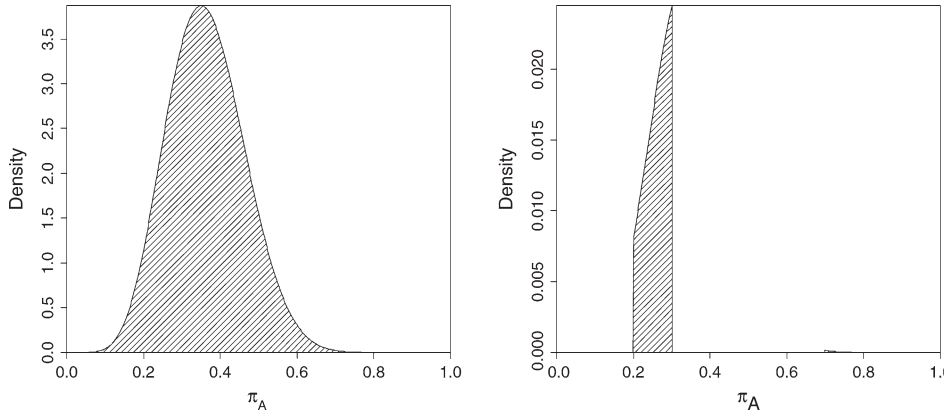


FIG. 2. Possible posterior densities for π_A.

Figure 1 shows two possible prior pdfs for π_A. The pdf on the left of Fig. 1, represents the previously given example where there is a 10% chance that π_A is less than 0.1, a 10% chance that π_A is greater than 0.9 and an 80% chance that π_A is between 0.1 and 0.9. The pdf on the right of Fig. 1 represents the prior belief that there is a 50% chance that π_A is between 0.2 and 0.3 and a 50% chance that π_A is between 0.7 and 0.8. Both the curves in Fig. 1 are pdfs and both obey the given probability statements. However, they are obviously very different.

Assume now that seven copies of allele A have been observed in the genotypes of 10 people so that the estimate from the data of π_A is 0.35. Figure 2 shows the effect of the prior pdfs in Fig. 1 on the posterior pdfs for π_A. The pdf on the right-hand side of Fig. 2 says that there is a high probability that π_A is between 0.2 and 0.3 and a very low probability that π_A is between 0.7 and 0.8. There is no chance that π_A is 0.35 even though this value has been observed in a sample of 10 people! The point being made here is that while both of these pdfs in Fig. 1 can be used to represent the scientist's prior beliefs about π_A, the pdf on the right-hand side of Fig. 1 will have a significant effect leading to the posterior density function which cannot explain the data. On the other hand, the pdf on the left-hand side of Fig. 1 will have only a minor effect in that it leads to a smooth curve which is centred very near to the estimate of π_A and decreases as one moves away from this.

Returning to the allele problem, one might think that a good choice of prior should be based on other samples or databases. For example, the scientist may say that it is known that the proportion is 0.53 in Aborigines, so it might be similar in Caucasians (assuming that the probability in the Caucasian population is the probability of interest). Alternatively, perhaps the scientist might say the proportion is 0.53 in Aborigines, 0.48 in Asians and 0.55 in Eastern Polynesians, therefore, its probable value might be somewhere in between these values for Caucasians. In another scenario, the scientists might say instead that they do not know anything except that π_A is probably between zero and one and after that fact all values of π_A are equally likely. Prior density functions can be chosen so that that these probability statements are represented statistically. This choice, however, is subjective—meaning it is the choice of the user and not dictated by an impartial/objective framework. It is this subjectivity that attracts substantial criticism from the detractors of Bayesian methods, who misguidedly cite that as a reason to dismiss the Bayesian approach.

Statisticians working with practical Bayesian problems attempt to choose 'uninformative' priors. An easy way to think of uninformative priors is to regard them as having very flat pdf. For example,

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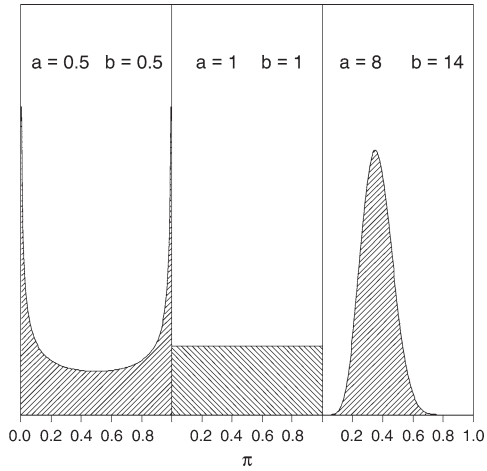


FIG. 3. The shape of the Beta distribution with respect to the choice of shape parameters *a* and *b*. The three graphs are not on the same vertical scale. The presentation here is for comparison of form only.

in a problem involving a prior on a mean, an uninformative prior might be a normal distribution with a mean of zero and a standard deviation of 1 million. Such a prior would say that the feasible range for the mean is somewhere between $+/-3$ million—a very broad range indeed. Uninformative priors are usually distributions that assign probabilities to a very broad range of values so that they do not have undue influence on the posterior distribution. In addition to having a 'deliberately vague' property, statisticians try to choose a prior from a family of pdfs with very well known properties that make the mathematics tractable. This second property has more to do with computational feasibility than statistical rigour.

In the case of a single proportion, such as the allele example being used here, a common choice is a family of curves called the Beta family. The shape of the Beta family is defined by two parameters, *a* and *b*. Some examples are given in Fig. 3.

It is plain to see from Fig. 3 that any choice of *a* and *b* that is different from 1 gives substantial shape to the curve. This shape will affect the posterior distribution and so leads to the criticism that if *a* and *b* are not 1, then the prior is *informative*, which is biased. Such criticism may be countered by allowing *a* and *b* to be 1, thereby choosing an *uninformative prior*.

4. Credible intervals versus confidence intervals

The assumption of a prior gives the user great power because it allows the derivation (analytically or computationally) of the posterior pdf of the parameter of interest. This pdf can then be used like any other pdf—to estimate means and variances (and any other moments of interest) and to answer probability questions. If the posterior density is continuous, then questions such as "what is the probability that π_A is 0.5?" cannot be answered because probability is equal to the area under the pdf. Therefore, "what is the probability that π_A is 0.5?" is mathematically equivalent to asking "what is the area under the curve between 0.5 and 0.5?" The answer of course is zero. Therefore, probability questions must be about ranges of values like, "what is the probability that π_A is between 0.1 and 0.9, i.e. what is Pr $(0.1 < \pi_A < 0.9)$?" The answer to this question and any like it is given by

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the area under the curve between the two values of the range. For example, the area under the curve between 0.1 and 0.9 is 0.9993873. This says that there is a 99.93873% chance that π_A is between 0.1 and 0.9. This logically leads to a natural subsequent question, namely, "for a specified probability p, what are the values of points l and u such that the probability that π_A lies between l and u is p?" Mathematically we express this as 'given p find l and u such that Pr $(l < \pi_A < u) = p$'. For example, one may ask "if $p = 0.95$, then what are l and u?" If the end-points l and u are determined using a posterior density function, then they define a credible interval for the parameter of interest. Assume, e.g. that the values of l and u for the current example are 0.18 and 0.57 for $p = 0.95$. This allows the forensic scientist to say that that the probability that π_A is between 0.18 and 0.57 is 0.95, or alternatively, that she is 95% sure that the true value of π_A is between 0.18 and 0.57. This may seem to the reader to be very much like a confidence interval. However, one would not be able to make such a statement with a confidence interval. Instead, the scientist would have to say she is 95% *confident* that π_A lies between 0.18 and 0.57. The word confident has a special meaning in this context. It can be read as "on average, 95% of intervals generated in the same way would contain the true value π_A". This is a fundamental difference.

It may not be immediately obvious, but there is an infinite set of intervals that satisfy Pr$(l < \pi_A < u) = p$, for a given value of p. For example, if the distribution under consideration is the Beta(1,1) or the uniform distribution, then choosing $l = 0,\ u = 0.95; l = 0.025,\ u = 0.975$ and $l = 0.05,\ u = 1$ all result in intervals with a probability of 0.95. The choice of where to start accumulating probability is somewhat arbitrary. HPD intervals attempt to capture the area of HPD—the most probable values, if you like—so that the resulting interval is as narrow as possible. Credible intervals on the other hand are chosen so that the central $100p\%$ of values lie between the end-points. More specifically, for a given value $p, l = q(\frac{1}{2}p)$ and $u = q(1 - \frac{1}{2}p)$, where $q(\alpha) = F^{-1}(\alpha)$ and $F^{-1}(\cdot)$ is called the inverse cdf. $q(\alpha)$ is usually referred to as the α quantile and is chosen so that the area under the pdf for values of less than $q(\alpha)$ is α. Many readers may be familiar with the concept of a percentile. A percentile is a quantile with the value of α multiplied by 100. For example, the 90th percentile is the 0.9 quantile.

Credible intervals and HPD intervals for most of the sampling error problem considered in this paper will be almost identical because of the symmetricity of the resulting posterior densities (on a logarithmic scale).

5. Credible intervals for LRs

One of the very useful properties of the Beta distribution is that if a Beta(a, b) prior distribution is assumed for a proportion and x alleles of type A alleles have been observed in a sample of N individuals ($2N$ alleles), then it can be shown that the posterior distribution is a Beta distribution with parameters $x + a$ and $2N - x + b$, i.e. Beta$(x + a,\ 2N - x + b)$. For example, if the scientist assumes a Uniform prior ($a = b = 1$) and observes seven copies of allele A in 10 individuals, then the resulting posterior distribution is Beta$(8, 14)$[2].

Figure 4 shows the three parts of Bayes Theorem. The prior density on π_A is a Beta$(1, 1)$, which means it takes value 1 for all values of π_A. Recall that there was an assumption of binomial sampling. This implies that the likelihood function for π_A given the data is the binomial distribution function. As noted at the start of this section the posterior density in this situation is Beta$(8, 14)$.

[2] See Appendix 1 for further details.

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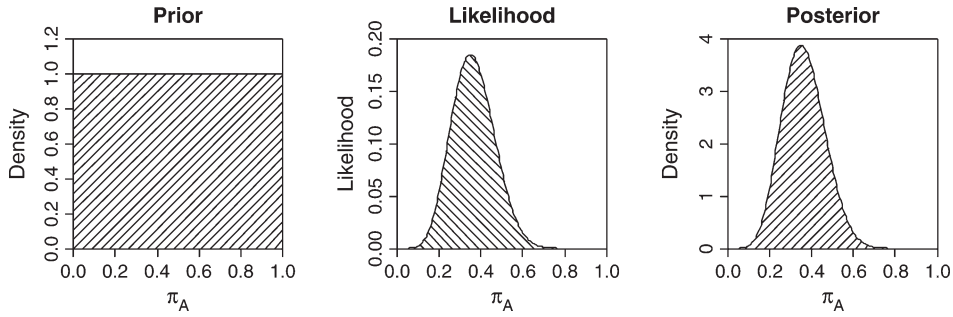


FIG. 4. Prior, likelihood and posterior distributions for the allele probability example.

It can be shown[3] that the posterior density differs from the likelihood by a constant. This explains the difference in the heights of the posterior density and the likelihood function. Uniform priors, such as the prior in Fig. 4 are used throughout this paper for ease of explanation. Forensic scientists generally feel more comfortable with the uniform prior rather than the more technically accurate reference prior (where both the parameters are ½). Furthermore, it should be noted that the method described in this paper is relatively insensitive to the choice of prior when the database is large ($N > 1000$).

The posterior density has a peak around 0.4 and nearly all the area under the curve is between about 0.1 and 0.8. In Section 4 it was shown that this posterior density can be used to derive a HPD interval or credible interval. This interval is a credible interval for the probability of a single allele. What remains to be explained is how this method helps assess sampling error in an LR calculation.

Assume that the scientist wishes to know the frequency of allele A at the LDLR locus for a case where a blood stain had been found at a crime scene. A suspect was apprehended in connection with the crime. DNA typing revealed the genotype of the crime scene stain at locus LDLR to be AB and also that the suspect was of type AB at this locus. Therefore, the suspect cannot be excluded and it is necessary to calculate some statistic that tells the court how compelling the evidence is against the suspect. The traditional approach is to calculate a match probability. However, many commentators argue that all evidence should be reported in the form of an LR. In this simple case there is a one-to-one correspondence mathematically between the LR and the match probability[4], P, so readers may choose whichever suits them. More formally, the LR compares the probability of the evidence, E, under the hypothesis H_p, that the suspect was the contributor of the DNA, to the probability of the evidence under the hypothesis H_d, that someone unrelated to the suspect was the contributor of the evidence. These hypotheses are not exhaustive and there may be others. Therefore, the LR is defined as

$$\mathrm{LR} = \frac{\Pr(E|H_p)}{\Pr(E|H_d)}. \tag{4}$$

[3] See Appendix 1.
[4] $\mathrm{LR} = \frac{1}{P}$.

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Using the substructure model of Balding & Nichols (1994) or recommendation 4.2 from the NRC, (4) becomes

$$\mathrm{LR} = \frac{(1+\theta)(1+2\theta)}{2[\theta+(1-\theta)\pi_{\mathrm{A}}][\theta+(1-\theta)\pi_{\mathrm{B}}]}. \tag{5}$$

When population substructure is not present ($\theta = 0$) expression (5) simplifies to

$$\mathrm{LR} = \frac{1}{2\pi_{\mathrm{A}}\pi_{\mathrm{B}}} \tag{6}$$

In order to derive an exact credible interval for expression (6) it would be necessary to derive the pdf, which is difficult even for a simple function such as this. However, because the posterior distribution of π_{A} is known, it is possible to find the end-points of the credible interval by simulation. If the posterior distribution of π_{A} is used to generate a random value of π_{A}, then one can imagine that this is similar to taking another sample (another database) and using that sample to estimate π_{A}. This estimate can in turn be 'plugged' into expression (6) to get an estimate of the LR. If this is done many times, the resulting sample of estimates of the LR can be used to estimate the end-points of a credible interval. This Monte Carlo simulation method has two advantages over the bootstrap approach. Firstly, the method is very fast because it requires only random number generation to obtain a set of allele frequencies and calculation of the LR. By comparison, the bootstrap method has three phases of computation: (1) a sample with replacement is taken from the database, which requires a medium level of computational effort; (2) the allele frequencies must be calculated by cross tabulation of the sample, which requires significant computational effort and (3) the LR is calculated. The second advantage of this Bayesian method is that it lends itself to a much more natural way of explaining the resulting interval.

Assume that a forensic laboratory has a one-locus testing kit with alleles A and B as above. The laboratory has a database with 7 alleles of type A and 13 alleles of type B. By assuming a uniform prior on the probability of allele A and assuming binomial sampling, then using Bayes Theorem it has been shown that the probability of allele A has a Beta(8, 14) distribution. Furthermore, assume the circumstances of the crime are given as above. Using the database, the estimate of the LR is $1/2 \times 0.35 \times 0.65 = 2.197802$. A 'new' value of π_{A} is then drawn from the posterior distribution. The first iteration yields $\pi_{\mathrm{A}} = 0.3931586$ and $\pi_{\mathrm{B}} = 1 - \pi_{\mathrm{A}} = 0.6068414$. Using these values, the LR is 2.09569. This procedure is repeated a further 99 times to get 100 estimated LRs, which have been sorted into ascending order below.

The 2.5th and 97.5th sample percentiles are estimates of the 0.025 and 0.975 quantiles of the distribution of the LR. Therefore, these points define a 95% credible interval for the LR in this case. From the data in Table 1, the end-points are halfway between the 2nd and 3rd smallest LRs and 2nd and 3rd largest LRs. Using the real data (which has slightly more precision than shown in Table 1), these values are 2.000803 and 3.124210. Therefore, the forensic scientist may say that she is 95% sure that the true LR is between 2.0 and 3.12 (with rounding).

6. Multiple alleles and multiple loci

The extension of this theory from a two-allele locus to a *k*-allele locus essentially involves using multivariate analogues of the results given thus far. The Beta distribution has a natural multivariate analogue called the Dirichlet distribution. The Dirichlet distribution allows the modelling of a whole

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TABLE 1 *This table contains* 100 *simulated LRs. The LRs are calculated using values of* π_A *drawn from the posterior distribution of* π_A *given the data. For example, when a value of* $\pi_A = 0.5$ *is drawn, then* $\pi_B = 1 - \pi_A = 0.5$ *and the LR is evaluated as* $1/(2 \times 0.5 \times 0.5) = 2$. *Instances where different values of* π_A *appear to yield the same LR are simply due to rounding errors. For example, where* $\pi_A = 0.500$ *and* $\pi_A = 0.503$, *the LRs are* 2 *and* 2.000072, *respectively*

π_A	π_B	LR	π_A	π_B	LR	π_A	π_B	LR	π_A	π_B	LR
0.500	0.500	2.000	0.400	0.600	2.084	0.361	0.639	2.168	0.307	0.693	2.352
0.503	0.497	2.000	0.601	0.399	2.085	0.360	0.640	2.170	0.295	0.705	2.402
0.493	0.507	2.000	0.602	0.398	2.087	0.360	0.640	2.170	0.293	0.707	2.415
0.512	0.488	2.001	0.398	0.602	2.088	0.359	0.641	2.173	0.288	0.712	2.440
0.486	0.514	2.002	0.394	0.606	2.094	0.359	0.641	2.173	0.287	0.713	2.442
0.486	0.514	2.002	0.607	0.393	2.095	0.357	0.643	2.179	0.283	0.717	2.462
0.475	0.525	2.005	0.393	0.607	2.096	0.351	0.649	2.195	0.281	0.719	2.473
0.474	0.526	2.005	0.393	0.607	2.096	0.351	0.649	2.196	0.281	0.719	2.474
0.471	0.529	2.007	0.391	0.609	2.099	0.350	0.650	2.198	0.270	0.730	2.536
0.461	0.539	2.012	0.390	0.610	2.102	0.349	0.651	2.200	0.267	0.733	2.553
0.456	0.544	2.015	0.389	0.611	2.103	0.348	0.652	2.204	0.267	0.733	2.554
0.554	0.446	2.024	0.388	0.612	2.106	0.346	0.654	2.211	0.258	0.742	2.612
0.439	0.561	2.031	0.387	0.613	2.108	0.345	0.655	2.211	0.257	0.743	2.615
0.438	0.562	2.031	0.386	0.614	2.109	0.342	0.658	2.221	0.257	0.743	2.618
0.436	0.564	2.034	0.386	0.614	2.109	0.339	0.661	2.232	0.254	0.746	2.640
0.431	0.569	2.039	0.382	0.618	2.118	0.330	0.670	2.262	0.253	0.747	2.645
0.424	0.576	2.047	0.380	0.620	2.123	0.325	0.675	2.278	0.249	0.751	2.676
0.418	0.582	2.055	0.377	0.623	2.129	0.325	0.675	2.279	0.227	0.773	2.849
0.416	0.584	2.059	0.377	0.623	2.130	0.316	0.684	2.312	0.224	0.776	2.876
0.415	0.585	2.060	0.371	0.629	2.142	0.316	0.684	2.313	0.220	0.780	2.911
0.407	0.593	2.072	0.371	0.629	2.142	0.314	0.686	2.321	0.212	0.788	2.993
0.406	0.594	2.073	0.369	0.631	2.148	0.314	0.686	2.321	0.206	0.794	3.059
0.403	0.597	2.078	0.368	0.632	2.149	0.313	0.687	2.327	0.195	0.805	3.183
0.401	0.599	2.082	0.633	0.367	2.151	0.311	0.689	2.334	0.193	0.807	3.213
0.400	0.600	2.083	0.363	0.637	2.162	0.307	0.693	2.349	0.186	0.814	3.300

set of proportions that add up to one instead of just a single proportion. To extend this procedure to multiple loci is trivial. Instead of sampling from one Dirichlet distribution in each iteration of the simulation process, samples are drawn from L Dirichlet distributions which correspond to the posterior density of allele probabilities at the lth locus. The resulting sample can then be used to estimate the LR. The details of this are contained in Appendix 2.

7. Conclusion

This paper addresses the problem of assessing sampling error in DNA calculations using a Bayesian approach. The Bayesian credible interval resulting from the method outlined in this paper allows the analyst to state, "I have assessed the statistical sampling error in this case. Based on my calculations I am 99% sure that the true LR lies between 550 million and 2 billion." The figures used in this statement have been chosen as an example. Some authors advocate the presentation of a lower bound only as to avoid confusing the court by presenting too many numbers. If this were the case, the

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statement would change to something like "Based on my calculations I am 99.5[5]% sure that the true LR lies is greater than 550 million."

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Appendix A

A.1 *The constant of proportionality in Bayes Theorem*

To achieve equality (rather than proportionality) in Bayes Theorem, the constant of proportionality must be assessed. The constant of proportionality in this case is the probability of the data, Pr(data). This probability is assessed by finding the marginal density by integrating the joint density over the parameter space. Bayes Theorem can then be expressed as

$$\Pr(\theta|\text{data}) = \frac{\Pr(\text{data}|\theta)\Pr(\theta)}{\int \Pr(\text{data}|\theta)\Pr(\theta)\,\mathrm{d}\theta}. \tag{A.1}$$

Equation (A.1) may be sometimes expressed in terms of pdfs instead of probabilities.

A.2 *Deriving the posterior density for a binomial proportion*

Recall that there was an assumption of binomial sampling. This means that the likelihood function for π_A is simply the binomial distribution function, i.e.

$$l(\pi_A|x, N) = \binom{2N}{x}\pi_A^x(1-\pi_A)^{2N-x} = \frac{(2N)!}{x!(2N-x)!}\pi_A^x(1-\pi_A)^{2N-x}. \tag{A.2}$$

[5] With a 99% credible interval, 0.5% of estimates will be below the lower bound and 0.5% of the estimates above the upper bound. If a one-sided credible interval is reported using the lower bound from a 99% two-sided credible interval then it becomes a 99.5% credible interval. This is because 0.5% of estimates will fall below the lower bound and 99.5% of estimates will exceed the lower bound.

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As previously noted, the posterior density in this situation is Beta$(x+1, 2N-x+1)$. The density function for the Beta distribution with these parameters is

$$\frac{1}{\beta(x+1, 2N-x+1)}\pi_{\mathrm{A}}^{x}(1-\pi_{\mathrm{A}})^{2N-x}, \tag{A.3}$$

where $\beta(a, b)$ is the complete Beta function. For integer arguments

$$\beta(a+1, b+1) = \frac{(a+b+1)!}{a!b!}, \tag{A.4}$$

so it can be shown that (A.3) becomes

$$\frac{(2N+1)!}{x!(2N-x)!}\pi_{\mathrm{A}}^{x}(1-\pi_{\mathrm{A}})^{2N-x}, \tag{A.5}$$

which in turn means that (A.5) differs from (A.2) by a factor of $2N+1$ or 21 in the example being used. This explains the difference in the heights of the posterior density and the likelihood function.

Appendix B

To extend this theory from a two-allele locus to a k-allele locus requires the use of a multivariate analogue of the Beta distribution, the Dirichlet distribution. The Dirichlet distributions can be used to model a set of proportions which are constrained to sum to one. More specifically, the probabilities of k alleles, which are denoted as π_i, for $i = 1, .., k$, can be modelled with a Dirichlet distribution such that $(\pi_1, \pi_2, \ldots, \pi_{k-1}) \sim$ Dirichlet$(\alpha_1, \alpha_{2,} \ldots, \alpha_k)$. Notice that there the subscripts on the π_i's only go up to $k-1$. This is because $\pi_k = 1 - \sum_{i=1}^{k-1} \pi_i$. When $k = 2$, the Dirichlet distribution is equivalent to a Beta distribution, i.e.

$$\pi_{\mathrm{A}} \sim \text{Dirichlet}(\alpha_1, \alpha_2) \equiv \pi_{\mathrm{A}} \sim \text{Beta}(\alpha_1, \alpha_2).$$

The Dirichlet family of distributions are conjugate priors for multinomial samples, in the same way that the Beta family of distributions are conjugate priors for binomial samples. This means that if a Dirichlet prior is assumed for the allele frequencies, then the posterior distribution of the allele frequencies also has a Dirichlet distribution. For example, assume we choose a uniform prior. This is a Dirichlet prior where $\alpha_i = 1$, for all k alleles. Now say we observe counts of x_i alleles of type A_i in a sample of $2N$ alleles, then the posterior density for $(\pi_1, \pi_2, \ldots, \pi_{k-1})$ is Dirichlet$\left(x_1+1, x_2+1, \ldots, 2N - \sum_{i=1}^{k-1} x_i + 1\right)$. To extend this procedure to multiple loci is trivial. Instead of sampling from one Dirichlet distribution in each iteration of the simulation process, samples are drawn from L Dirichlet distributions which correspond to the posterior density of allele probabilities at the lth locus. The resulting sample can then be used to estimate the LR and a credible interval for the LR.

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